INCH-POUND

MIL-DTL-24282A(SH)
14 July 2026
SUPERSEDING
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3 July 1968

DETAIL SPECIFICATION

LIGHT, SIGNAL, MULTIPURPOSE, KIT, NONMAGNETIC

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a nonmagnetic, portable, multipurpose light kit, complete with a stowage box for signaling, visual communications, and limited lighting for shipboard use. It can be operated by 115-volt alternating current (AC) or by self-contained dry cell batteries.

1.2 Classification. Multipurpose signal lights are of the following types (see 6.1):

a. Type I - Incandescent (Sym 106.1)

b. Type II - Light emitting diode (LED) replacement light assembly

c. Type III - Solid state lighting (SSL)

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	-	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)
TT-P-1757	-	Primer Coating, ALKYD Base, One Component
W-C-596/14	-	Connector, Cable Outlet, Electrical, General Purpose, Hospital Grade, Cable Connecting, Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz
W-C-596/140	-	Connector, Receptacle, Electrical, Male Inlet, General Purpose, Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz

FEDERAL STANDARDS

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 5850

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FED-STD-H28 - Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTIONS

A-A-59262 - Battery, Dry, BA-2

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-900 - Rubber Gasket Material, 45 Durometer Hardness

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-2726/38 - Receptacle, Plug, Electrical, 15-Ampere, 125-Volt, Bladed Type, Grounded (Symbol Number 1218.3)

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-DTL-15090 - Enamel, Equipment, Light Gray (Navy Formula No. 111)

MS16569 - Switch, Toggle, Double Pole, Single Throw (for Use in Lighting Fixtures)

MIL-PRF-22684/8 - Resistor, Fixed, Film, Insulated, Style RL42….TX

MIL-I-24768/1 - Insulation, Plastic, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin (GME)

MS25100 - Switch, Toggle, Two Pole, Sealed Toggle (inactive for new design)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-202-301 - Test Method Standard, Method 301, Dielectric Withstanding Voltage

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

MIL-STD-1353 - Electrical Connectors, Plug-in Sockets and Associated Hardware, Selection and Use of

MIL-STD-1399-300-1 - Department of Defense Interface Standard Section 300, Part 1 Low Voltage Electric Power, Alternating Current

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454 - General Guidelines for Electronic Equipment

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C78.377 - Electric Lamps – Specifications for the Chromaticity of Solid-State Lighting (SSL) Products

(Copies of this publication are available online at https://webstore.ansi.org.)

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AMERICAN WELDING SOCIETY (AWS)

AWS WHB-1.10 - Welding Handbook Volume 1 – Welding and Cutting Science and Technology

(Copies of this publication are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM A342/A342M - Standard Test Methods for Permeability of Weakly Magnetic Materials

ASTM D5948 - Standard Specification for Molding Compounds, Thermosetting

(Copies of these publications are available online at https://www.astm.org.)

ILLUMINATING ENGINEERING SOCIETY OF NORTH AMERICA (IES)

ANSI/IES LM-79 - Approved Method: Optical and Electrical Measurements of Solid-State Lighting Products

(Copies of this publication are available online at www.ies.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code

(Copies of this publication are available online at https://www.nfpa.org.)

SAE INTERNATIONAL

SAE AS7928 - Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification For

SAE AS31021 - Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type

SAE AS31061 - Connector, Plug, Electric, Straight, Solder Contacts, AN Type

SAE J578 - Chromaticity Requirements for Ground Vehicle Lamps and Lighting Equipment

(Copies of these publications are available online at https://www.sae.org.)

UNDERWRITERS LABORATORIES (UL)

UL 62 - Flexible Cords and Cables

(Copies of this publication are available online at https://www.shopulstandards.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design. The light shall be suitable for signaling, visual communications, and limited lighting, and shall be examined as specified in 4.3.3.

3.3 Construction. The signal light shall be constructed for long service without frequent overhaul under severe conditions encountered aboard ship. The construction shall be simple and sturdy and such that continued satisfactory operation in accordance with this specification and routine maintenance in service do not require special tools or excessively precise workmanship. Special tools are defined as those tools not listed in the GSA Global Supply Catalog. (Copies of this Catalog found with the Defense Contract Management Agency [DCMA] at https://cmls.gsa.gov.) The construction shall be examined as specified in 4.3.3.

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3.4 Materials. Materials shall be as specified herein and shall meet the performance requirements specified herein. All materials shall be non-corrosive and nonmagnetic.

3.4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The materials shall be examined as specified in 4.3.3.

3.4.2 Magnetic permeability. The magnetic permeability of all fixture parts shall be 2.0, or less, after fabrication and this shall be verified by magnetic permeability testing as specified in 4.4.7. The transformer, if used, is excluded.

3.4.3 Plastics. Plastic materials shall be free of defects such as embedded particles, bubbles, scratches, blemishes, and striations and shall be examined as specified in 4.3.3.

3.4.4 Stress relief. Stress relief methods shall be examined as specified in 4.3.3.

3.4.4.1 Metals. All metals used in fabrication and assembly shall be treated to prevent distortion or failure due to stresses or other conditions resulting from working, forming, and welding. Heating or peening after welding will accomplish stress relief in accordance with AWS WHB-1.10.

3.4.4.2 Plastics. Adequate measures shall be taken in molding and processing plastics to ensure that stress buildup does not occur or is satisfactorily treated to relieve these stresses to prevent deterioration or failure of the part or assembly.

3.4.5 Insulating materials. Insulating material shall be type GME in accordance with MIL-I-24768/1. Molding compounds shall be in accordance with ASTM D5948. The insulating material shall be examined as specified in 4.3.3.

3.4.6 Rubber. Material for rubber gaskets shall be in accordance with MIL-PRF-900. The material shall be examined as specified in 4.3.3.

3.4.7 Dissimilar metals. Application and use of dissimilar metals shall be in accordance with the requirements of MIL-DTL-917. The material shall be examined as specified in 4.3.3.

3.4.8 Restricted material. Material listed in MIL-DTL-917 as prohibited shall not be used.

3.5 Threaded parts. Screw threads shall be Unified National Coarse (UNC) class 2 in accordance with FED-STD-H28 and verified in accordance with 4.3.3.

3.5.1 Inserts, tapping, and threaded devices. Inserts, tapping, and threaded devices shall be as follows:

a. Self-tapping screws and sheet metal screws shall not be used.

b. Molded-in-place inserts in plastic for screw threads are required.

3.5.2 Locking devices. Locking devices shall be used to secure mechanical assemblies to provide continuous operation under the conditions of shock and vibration specified herein.

3.6 Welding. Welding and allied processes used in fabrication shall be in accordance with AWS WHB-1.10 and verified in accordance with 4.3.3.

3.7 Castings. Castings shall be free from cold shuts, blow holes, or any imperfections that may affect strength. They shall be finished smooth, that is, no openings or rough edges. Verification shall be in accordance with 4.3.3.

3.8 Dimensional tolerances. When a figure or paragraph expresses a dimension in this document, the following tolerances shall apply (verification shall be in accordance with 4.3.3):

a. Fractional ±1/64 inch

b. Decimal ±0.005 inch

c. Angular ±1 degree

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3.9 Electrical requirements.

3.9.1 Clearances. Minimum electrical creepage and clearances shall be as required by NFPA 70. Verification shall be in accordance with 4.3.3.

3.9.2 Terminals. Terminals shall be in accordance with SAE AS7928. Terminals shall be used on all inside light connections. Verification shall be in accordance with 4.3.3.

3.9.3 Exposed metal or other conductive parts. Design and construction of the equipment shall be so that all exposed parts, panels of metal, or other electrically conductive material are at the ship's receptacle's ground potential when operated via the power cable. The light assembly and the stowage box shall be grounded to the receptacle's ground via the green conductor of the power and extension cables. Means shall be provided to attach the green conductor of the cable to bare metal on the inner face of the light assembly housing and stowage box. Precautions shall be taken to ensure the safety of operational personnel to the maximum practicable extent. Verification shall be in accordance 4.4.9, with no impedance measured at greater than 0.10 ohm.

3.9.4 Safety. Precautions shall be taken to ensure the safety of operating personnel. When the light assembly is connected properly to the case and the case is grounded through the power cable, there shall be no accessible way for operating personnel to receive an electric shock even though an internal fault may exist between any circuit and ground or any circuit and the light or case enclosure. The design shall hold to a practical minimum the possibility of maintenance personnel being exposed to electric shock while servicing, adjusting, or checking out the fixtures or parts. Verification shall be in accordance with 4.3.3.

3.9.5 Shipboard power interface. The signal light shall operate on ungrounded 115V type I power as described in MIL-STD-1399-300-1. Verification shall be in accordance with 4.4.1.

3.9.6 Electromagnetic emissions. The signal light shall meet electromagnetic emissions requirements in accordance with MIL-STD-461 RE102 surface ship above deck limits. Verification shall be in accordance with 4.4.11.

3.10 Performance characteristics.

3.10.1 Operation. The signaling light, when tested as specified in 4.4.1, shall produce all required light outputs.

3.10.2 Dielectric withstanding voltage. There shall be no evidence of breakdown when the light assembly is subjected to the test specified in 4.4.2.

3.10.3 Insulation resistance. The insulation resistance between all current-carrying and noncurrent-carrying parts shall be not less than 1 megohm (see 4.4.3).

3.10.4 Beam spread. The signal light shall be designed to permit the following beam spread limits:

Horizontal – 7 degrees
Vertical – 7 degrees

3.11 Environmental conditions.

3.11.1 Shock. Stowage box, signal light, and components shall meet shock requirements for grade A, class I, type A equipment in accordance with MIL-DTL-901 and as specified in 4.4.4.

3.11.2 Vibration. Stowage box, signal light, and components shall meet type I environmental vibration requirements in accordance with MIL-STD-167-1 and as specified in 4.4.5.

3.12 Light assembly capability. The multipurpose portable signal light shall contain a light assembly that produces a high-intensity beam of light suitable for use as a spotlight or as a blinker for visual communications when the beam is electrically interrupted by means of a trigger switch. The light assembly shall provide for sighting and holding the beam on the desired target. The light assembly shall be designed to permit wireless operation from self-contained batteries (see 3.13.6) or from a power source within the stowage box via the extension cable (see 3.13.7.2). The capability shall be verified in accordance with the operational testing of 4.4.1.

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3.13 Components.

3.13.1 Kit. Each kit shall contain one stowage box (see 3.13.7), which, in turn, shall contain the following:

a. Complete light assembly (see 3.13.2).

b. Lenses, one each of red, green, and yellow (see 3.13.2.1.8). Required for type I and type II and if necessary for type III.

c. Extension cable (see 3.13.3).

d. Power cable (see 3.13.4).

e. Instruction manual (see 3.13.5 and 6.2).

Verification of the kit components shall be in accordance with 4.3.3.

3.13.2 Light assembly capability. The multipurpose, portable signal light shall contain a light assembly that produces a high-intensity beam of light suitable for use as a spotlight or as a blinker for visual communications when the beam is electrically interrupted by means of a trigger switch. The light assembly shall provide for sighting and holding the beam on the desired target. The light assembly shall be designed to permit wireless operation from self-contained batteries (see 3.13.6) or from a power source within the stowage box via the extension cable (see 3.13.7.2). The general dimensions, configurations, and arrangement of the equipment shall be as shown on figures 1 and 2 and as specified hereinafter.

3.13.2.1 Type I and type II light assembly. The general dimensions, configurations, and arrangement of the type I or type II light assembly shall be as shown on figures 1 and 2 and as specified hereinafter and verified by examination in accordance with 4.3.3.

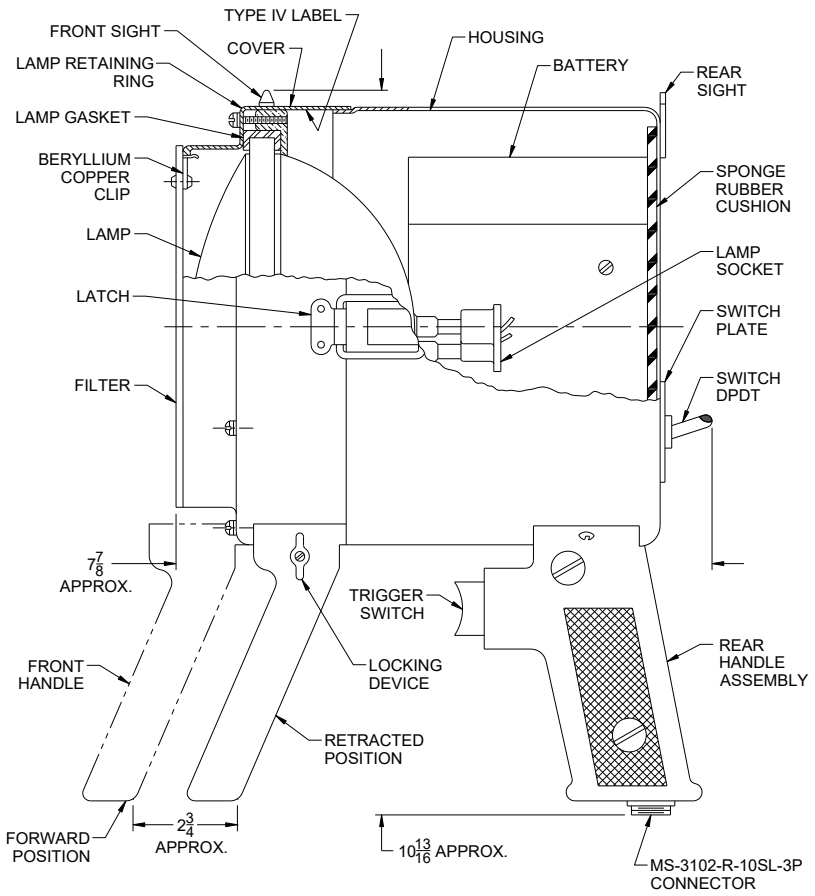


FIGURE 1. Type I and type II light assembly.

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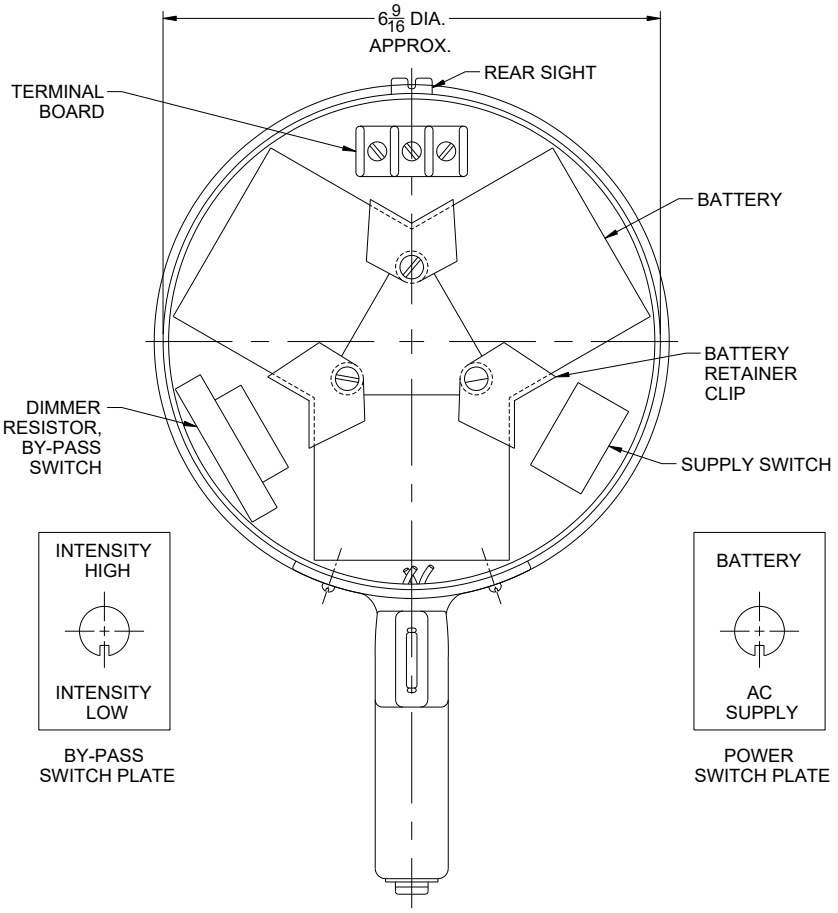


FIGURE 2. Type I and type II light assembly with cover removed.

3.13.2.1.1 Housing. The housing shall consist of an aluminum cylinder closed at one end. The housing shall be so designed as to accommodate three batteries (see 3.13.6), two switches (see 3.13.2.1.4), a rear handle (see 3.13.2.1.5.2), a cover (see 3.13.2.1.2), a rear sight (see 3.13.2.1.3), and the internal wiring and hardware specified in 3.13.2.1.6. Verification shall be in accordance with 4.3.3.

3.13.2.1.2 Cover. The aluminum cover shall be designed to accommodate the lens (see 3.13.2.1.8), light source (see 3.13.2.1.7), front handle (see 3.13.2.1.5.1), and front sight (see 3.13.2.1.3). The cover shall be firmly attached to the housing by means of two latches. Verification shall be in accordance with 4.3.3.

3.13.2.1.3 Sights. Sights shall be provided as shown on figures 1 and 2, and for type I shall be aligned to be parallel to the optical axis of the sealed beam lamp. The rear sight shall be a “V” notched type, and the front sight shall be the beaded type. Verification shall be in accordance with 4.3.3.

3.13.2.1.4 Switches. Switches shall be mounted on the rear of the housing, one switch to either side of a vertical centerline established between the rear sight and the rear handle. The exact location of each switch is at the manufacturer’s discretion. A toggle switch, double pole, double throw power transfer (ON-OFF-ON) type in accordance with MS25100 (inactive for new design), part number MS25100-23, shall be provided to permit switching between AC power from the stowage case and the direct current (DC) power of the batteries internal to the light assembly. A second toggle switch, single pole, single throw, in accordance with MS16569, part number MS16569-1, shall be used to switch between high (full) intensity and low intensity. Verification shall be in accordance with 4.3.3.

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3.13.2.1.5 Handles.

3.13.2.1.5.1 Front. The front handle shall be of aluminum, and shall be finished smooth, with no openings or rough edges. The handle shall be an adjustable type to ensure a steady position while signaling. A device shall be provided for locking in the forward position without the use of a tool. If necessary, the handle may be mounted off-center. Verification shall be in accordance with 4.3.3.

3.13.2.1.5.2 Rear. The rear handle (pistol grip) shall be made of either aluminum or molded plastic of a grade adapted for the purpose. The rear handle shall contain a trigger-operated switch capable of momentary contact such as would be required for signaling a light beam. The trigger switch shall have positive open and closed positions and, when operated at the rate specified in 4.4.6, it shall successfully complete 50,000 cycles of operation. A male receptacle, part number MS3102-R-10SL-3P in accordance with SAE AS31021, shall be securely mounted as shown on figure 1, and shall be connected as shown on figure 3. A protective cap for the receptacle shall be attached via a safety wire to the rear handle to cover the receptacle when the light assembly is used off internal batteries or stowed.

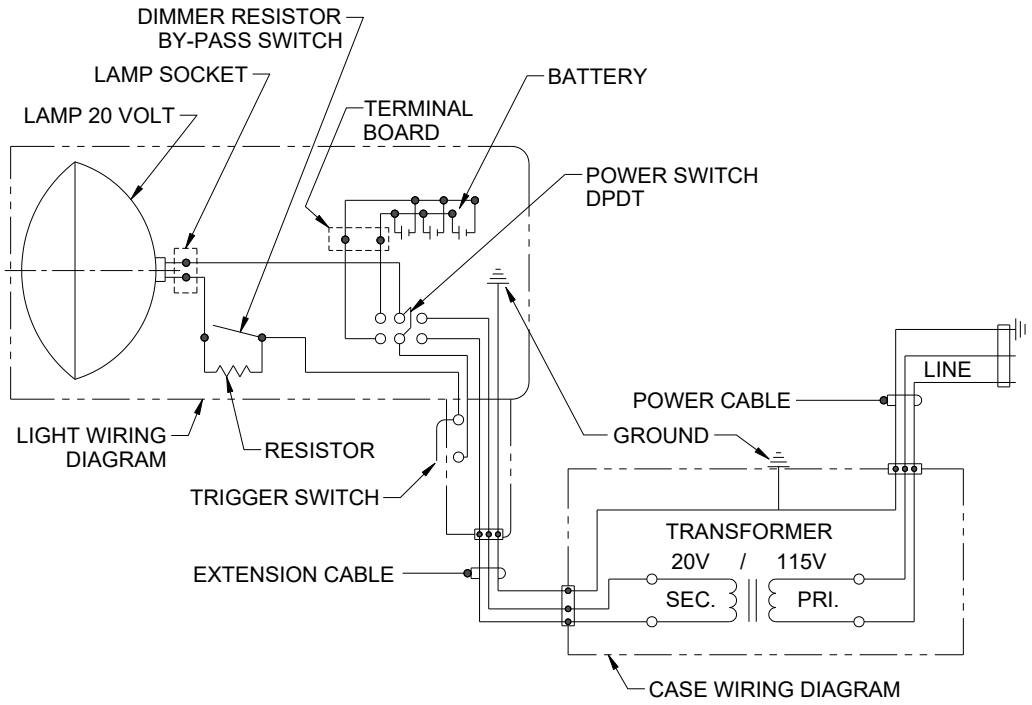


FIGURE 3. Type I wiring diagram.

3.13.2.1.6 Internal wiring. Internal wiring shall be a minimum of 18 AWG and be chosen in accordance with the hook-up wire table of MIL-DTL-917. For type I light assemblies, connection to the lamp shall be made by a mogul prong lampholder with a minimum rating of 250 volts and 75 watts. Verification shall be in accordance with 4.3.3.

3.13.2.1.7 Light source.

3.13.2.1.7.1 Type I lamp. The lamp shall be sealed beam, PAR-46, Industry Number 4300, 20-volt, 16,000-lumens. The lamp shall be fitted to a V-shaped rubber gasket and shall be firmly mounted to the cover by means of an easily removable retaining ring. Dimming shall be accomplished by placing a 2-watt, 10-ohm resistor, part number M22684/08-1002 in accordance with MIL-PRF-22684/8, in series with the lamp filament as shown on figure 3. Verification shall be in accordance with 4.3.3.

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3.13.2.1.7.2 Type II LED device. The type II light source shall be a device utilizing LEDs as the source of light and shall fit into the existing opening of the light assembly. The LED device color temperature shall fall within either the 2,700- or 3,000-Kelvin chromaticity quadrangles defined in ANSI C78.377. The LED device shall produce a minimum of 16,000 lumens at full brightness. The dimmed output shall fall between 40 and 60 percent of the measured light output at full brightness. The LED device shall contain all circuitry required to operate and meet light output requirements on both 20-volt alternating current (VAC) and 20-volt direct current (VDC) power. The maximum power draw of the LED device shall be 10 watts in any operating mode. Verification of the type II LED device's performance shall be in accordance with 4.4.8.

3.13.2.1.8 Lens. Lenses are used to filter the white light of either the incandescent lamp or LED device. The lenses shall be furnished in three colors (red, yellow, and green) in accordance with SAE J578. The lens shall consist of a plastic disc 5⅜ inches in diameter and 3/32 inch thick. The lens shall be easily attached to the cover by means of three copper-beryllium clips. Verification shall be in accordance with 4.3.3.

3.13.2.2 Type III light assembly.

3.13.2.2.1 Dimensions. The type III light assembly, when fully assembled, shall not exceed the length, width, or height of the type I/II light assembly. Verification shall be in accordance with 4.3.3.

3.13.2.2.2 Sights. Two sights shall be mounted on top of the assembly along the centerline of the housing to assist in aiming the signal light. The rear sight shall be a "V"-notched type, and the front shape shall be a beaded type. The sights shall be 0.25 inch or less in any dimension. Verification shall be in accordance with 4.3.3.

3.13.2.2.3 Switches. All switches shall be mounted on the rear face of the housing that faces the user when the light assembly is being aimed at a target. A minimum of two switches are required. One switch shall be used to change between the internal battery, the stowage box power source via extension cable (see 3.13.3), and OFF. A second switch shall switch between high and low intensities. Verification shall be in accordance with 4.3.3.

3.13.2.2.4 Handles. The light assembly shall have two handles to aid in aiming the light assembly.

3.13.2.2.4.1 Forward handle. The forward handle shall be adjustable so that its distance to the rear handle can be increased up to 3 inches forward from its furthest rear position. A method for locking the handle in both the forward and rear positions shall be provided. The ability to move the handle and lock it in place shall not require the use of tools. Verification shall be in accordance with 4.3.3.

3.13.2.2.4.2 Rear handle. The rear handle shall contain the following features:

a. Trigger switch. A trigger-operated switch that turns on the light beam when pressed and the light beam shall extinguish instantly when the trigger is released. The trigger switch shall be rated for a minimum of 50,000 cycles. The speed at which the trigger switch turns off and interrupts the light beam shall be suitable for communicating information via Morse code at a minimum rate of 12 words per minute. Verification shall be in accordance with 4.4.6.

b. Electrical connector. The rear handle shall allow for the connection of the extension cable (see 3.13.3) to the light assembly via an electrical connector selected in accordance with MIL-STD-1353. Cadmium-plated connectors shall not be used. Verification shall be in accordance with 4.3.3.

c. Attached protective cap. A protective cap for the receptacle shall be attached via a safety wire to the rear handle to cover the receptacle when the light assembly is used off internal batteries or stowed. Verification shall be in accordance with 4.3.3.

3.13.2.2.5 Internal wiring. Internal wiring shall be in accordance with the requirements of MIL-DTL-917. Verification shall be in accordance with 4.3.3.

3.13.2.2.6 Light source. The light source shall be a device using SSL technology. The device may contain additional integrated electrical components and circuitry required for operation. The device shall be field replaceable without having to replace the entire light assembly.

3.13.2.2.6.1 White light. The white light color temperature shall fall within either the 2,700- or 3,000-Kelvin chromaticity quadrangles defined in ANSI C78.377. The white light output shall be a minimum of 16,000 lumens at full brightness. Verification shall be in accordance with 4.4.8.

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3.13.2.2.6.2 Color light. A means for changing the white output to red, yellow, or green outputs shall be provided without the use of tools. The red, amber, and green colors shall fall within their respective quadrangles established in SAE J578. The measured red lumens shall be 20 to 30 percent of the measured white output at full brightness. The measured yellow lumens shall be 40 to 60 percent of the measured white output at full brightness. The measured green lumens shall be 30 to 50 percent of the measured white output at full brightness. Verification shall be in accordance with 4.4.8.

3.13.2.2.6.3 Dimmed output. The dimmed output of white light, and any color, shall fall between 40 to 60 percent of the respective measured light output at full brightness. Verification shall be in accordance with 4.4.8.

3.13.2.2.7 Access for maintenance. The light assembly shall provide access to remove the batteries and the light source device. Verification shall be in accordance with 4.3.3.

3.13.2.3 Enclosure effectiveness. The light assembly, when fully assembled, shall meet the spraytight requirement of MIL-STD-108. This requirement shall be met both with the extension cable (see 3.13.3) inserted into the assembly and with it removed and the receptacle covered with the attached cap. Verification shall be in accordance with 4.4.10.

3.13.3 Extension cable. A cable assembly for supplying power to the signal light from the stowage box shall be furnished. This cable shall be 25 feet long. Verification shall be in accordance with 4.3.3.

3.13.3.1 Type I/II extension cable. The extension cable for type I and type II signal lights shall consist of cable, type UL-62 SOOW 18/3 fitted with electrical connectors, part number MS3106-R-10SL-3S in accordance with SAE AS31061 at both ends.

3.13.3.2 Type III extension cable. The extension cable for type III signal lights shall meet the requirements of UL 62 and shall not utilize polyvinyl chloride (PVC) insulation. The cable shall have three conductors, minimum, with one conductor dedicated to providing a ground connection to the light assembly. The extension cable shall be terminated in connectors at both ends selected from MIL-STD-1353 and that mate with the connectors selected for the rear handle (see 3.13.2.2.4.2) and stowage box (see 3.13.7). Cadmium-plated connectors shall not be used.

3.13.4 Power cable. A cable assembly for supplying power to the stowage box from the ship's 115 VAC service receptacles shall be furnished. It shall be 15 feet long, type UL-62 SOOW 14/3, with a female plug, part number WC596/14-2, in accordance with W-C-596/14 at one end and a male plug, part number M2726/38-001, in accordance with MIL-DTL-2726/38, on the other. Verification shall be in accordance with 4.3.3.

3.13.5 Instruction manual. An instruction manual shall be furnished for each complete signal light. The required content of the instruction manual shall be as specified (see 6.2). Verification shall be in accordance with 4.3.3.

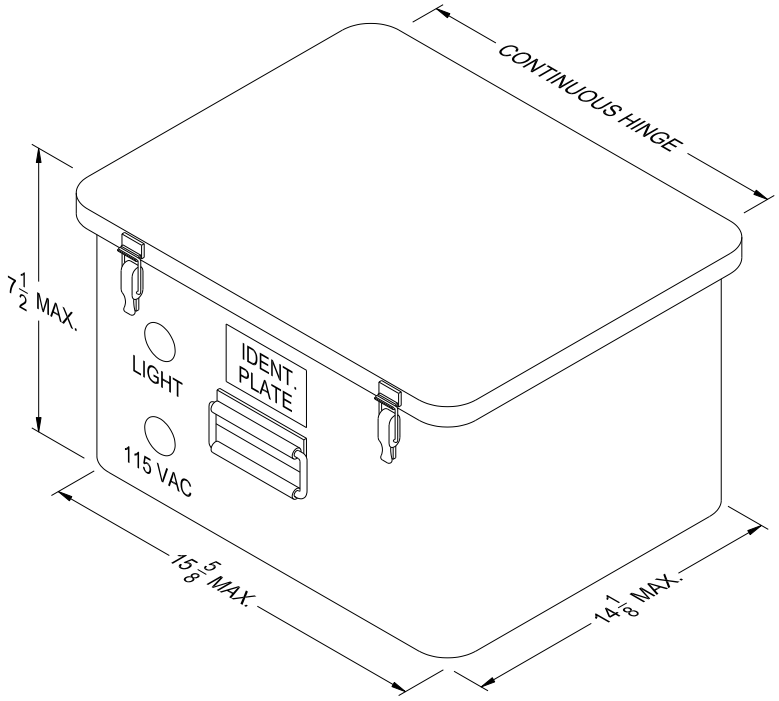
3.13.6 Batteries. The requirements for the battery or batteries used within the light assembly shall be as follows. Verification shall be in accordance with 4.3.3.

3.13.6.1 Batteries for type I and type II lights. Type I and type II light assemblies shall utilize three batteries (part number AA59262BA-2), 22½ volts in accordance with A-A-59262. The three batteries shall be connected in parallel. Unless otherwise specified (see 6.2), batteries need not be furnished by the manufacturer.

3.13.6.2 Batteries for type III lights. The battery or batteries for type III light assemblies shall be primary (non-rechargeable) and not exceed 30 VDC when installed. The batteries shall permit steady operation of the light assembly at high (full) intensity for a minimum of 1 hour. After 1 hour of continuous use, the total lumens of the white output at high (full) intensity shall be within 10 percent of the maximum lumens. The batteries shall be replaceable in the field using basic hand tools. Unless otherwise specified (see 6.2), the quantity, chemistry, size, and configuration of the batteries shall be as determined by the manufacturer to meet the performance requirements of this specification.

3.13.7 Stowage box. The stowage box shall not exceed the dimensions shown on figure 4. The material shall be a commercial grade aluminum that is protected against corrosion and of sufficient gage to provide firm support for component and compartmentation without excessive weight. A commercially available box and cover, adapted for the purpose, made of drawn or fabricated construction will be acceptable. Verification shall be in accordance with 4.3.3.

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MULTIPURPOSE SIGNAL LIGHT KIT
SYMBOL 106.1, MIL-DTL-24282
CONTRACT NO. _____ DATE (YEAR)
MANUFACTURER OR TRADEMARK

IDENTIFICATION PLATE

FIGURE 4. Type I and type II stowage box.

3.13.7.1 Compartmentation. Compartmentation shall be provided for items specified in 3.13.1. Aluminum baffles and foamed plastic, rubber, or felt cushions shall be provided to prevent movement and damage of components in transit and stowage aboard ship. Kits shall conform to the requirements of 3.11.1 and 3.11.2.

3.13.7.2 Stowage box power supply.

3.13.7.2.1 Type I and type II transformer. Type I and type II signal light stowage boxes shall provide approximately 20 VAC power to the light assembly via a transformer located within the stowage box. The transformer shall be rated at a minimum of 25 watts, with a primary voltage of 115 to 120 volts and a secondary voltage of 22 to 26 volts at 60 hertz. The transformer shall use integrated leads for conductor termination. The transformer shall be firmly mounted to the box and wired to receptacles as shown on the wiring diagram, figure 3.

3.13.7.2.2 Type III power supply. Type III signal light stowage boxes shall contain an internal power supply or transformer that operates from ship's 115 VAC service and outputs the power required by the type III light assembly. The voltage provided to the light assembly shall not exceed 30 volts. The power supply shall be firmly mounted within the stowage box and shall be electrically bonded to the ground conductor of the power cable in accordance with the requirements of MIL-DTL-917.

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3.13.7.3 Light assembly receptacle. The light assembly receptacle for the extension cable (see 3.13.3) shall be attached to the box on the same face as the power supply receptacle and as close to the transformer or power supply outputs as practicable. A plastic cap attached to a safety wire shall be anchored to the stowage box to provide protection for the receptacle with the signal light when it is not in use.

3.13.7.3.1 Type I and type II light assembly receptacle. The light assembly receptacle shall be part number MS3102-R-10SL-3P in accordance with SAE AS31021.

3.13.7.3.2 Type III light assembly receptacle. The light assembly receptacle shall be the same as the receptacle chosen for the rear handle (see 3.13.2.2.4.2).

3.13.7.4 Power supply receptacle. The power supply receptacle shall be part number WC596/140-1 in accordance with W-C-596/140 and shall be flush-mounted to the box adjacent to the transformer.

3.14 Weight. Component parts shall be as small and light as possible but sufficiently rugged to withstand rough handling so that mechanical damage or distortion will not be of sufficient magnitude as to impair continued operation. The total weight of the signal light shall not exceed 4¼ pounds without batteries. The total weight of the signal light kit, without batteries, shall not exceed 16½ pounds. Verification shall be in accordance with 4.3.3.

3.15 Identification and switch plates. Identification and switch plates shall be in accordance with MIL-DTL-15024. Identification plates shall be securely attached on the stowage box as shown on figure 4. The identification plate shall clearly display “MULTIPURPOSE SIGNAL LIGHT KIT”, “MIL-DTL-24282”, the date of manufacture, and the manufacturer’s name and CAGE code. Switch plates for type I and type II light assemblies shall be as shown on figure 2 and shall be securely attached on the signal light as shown on figures 1 and 2. Switch plates for type III light assemblies shall be provided for each switch, with the style to be at the manufacturer’s discretion. Verification shall be in accordance with 4.3.3.

3.16 Protection against corrosion. In order to prevent corrosion, all metal parts shall be constructed from corrosion-resisting materials or other materials treated in a manner to render them adequately resistant to corrosion. Treatment shall be in accordance with MIL-DTL-917. Verification shall be in accordance with 4.3.3.

3.17 Painting. All metal parts shall be completely painted in accordance with the requirements specified hereinafter. Plastic enclosures normally will not be painted. Verification shall be in accordance with 4.3.3.

3.17.1 Cleaning. After all machining, welding, and brazing operations are completed, the exterior and interior surfaces of all metal enclosures shall have all rust or other visible corrosion products removed. The exterior and interior surfaces shall be thoroughly cleaned of all grease, oil, and dirt by solvent wiping, vapor degreasing, or caustic washing and rinsing.

3.17.2 Undercoats. Either of the following undercoats shall be applied:

a. One coat of primer pretreatment coating, type III or type IV in accordance with TT-C-490, shall be applied as a continuous film 0.0002 to 0.0005 inch thick.

b. A hot diptank phosphate treatment, type I in accordance with of TT-C-490, followed by one coat of primer in accordance with TT-P-1757, and applied as a continuous film 0.0002 to 0.001 inch thick.

3.17.3 Finish coats. Light gray enamel, in accordance with MIL-DTL-15090, shall be applied over the undercoat at a thickness capable of providing protection of the equipment and an appearance meeting the requirements of 3.17.

3.18 Workmanship. The signaling light shall be manufactured and finished in a manner that is consistent with best industry practices. Particular attention shall be paid to neatness and thoroughness of soldering, crimping, wiring, marking of parts and assemblies, plating, painting, machine screw assemblage, welding, and freedom of parts from burrs and sharp edges. See the workmanship guideline of MIL-HDBK-454 for additional information. Verification shall be in accordance with 4.3.3.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified in table I are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

TABLE I. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Design	X	X	3.2	4.3.3
Construction	X	X	3.3	4.3.3
Recycled materials	X		3.4.1	4.3.3
Magnetic Permeability	X	X	3.4.2	4.4.7
Plastic defects	X	X	3.4.3	4.3.3
Stress relief	X		3.4.4	4.3.3
Other material characteristics	X		3.4.5, 3.4.6, 3.4.7	4.3.3
Restricted material	X		3.4.8	4.3.3
Threaded parts	X	X	3.5	4.3.3
Welding	X	X	3.6	4.3.3
Castings	X	X	3.7	4.3.3
Dimensional tolerances	X	X	3.8	4.3.3
Electrical	X		3.9.1, 3.9.2, 3.9.4	4.3.3
Electrical bonding	X		3.9.3	4.4.9
Power interface	X		3.9.5	4.4.1
Operation	X	X	3.10.1	4.4.1
Dielectric withstanding voltage	X	X	3.10.2	4.4.2
Insulation resistance	X	X	3.10.3	4.4.3
Beam spread	X		3.10.4	4.4.8, 4.4.8.1, 4.4.8.2
Shock	X		3.11.1	4.4.4
Vibration	X		3.11.2	4.4.5
Light assembly capability	X		3.12	4.4.1
Electromagnetic emissions	X		3.9.6	4.4.11
Kit components	X	X	3.13.1, 3.13.2, 3.13.3, 3.13.4, 3.13.5, 3.13.6, 3.13.7	4.3.3

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TABLE I. First article and conformance inspections - Continued.

Characteristic	First article	Conformance	Requirement	Verification
Type I and type II light assembly design	X		3.13.2.1.1 3.13.2.1.2 3.13.2.1.3 3.13.2.1.4 3.13.2.1.6 3.13.2.1.8	4.3.3
Type I and type II front handle design	X		3.13.2.1.5.1	4.3.3
Type I and type II rear handle design	X		3.13.2.1.5.2	4.4.6
Type I lamp	X	X	3.13.2.1.7.1	4.3.3
Type II LED device	X		3.13.2.1.7.2	4.4.8
Type III light assembly design	X		3.13.2.2.1 3.13.2.2.2 3.13.2.2.3 3.13.2.2.5 3.13.2.2.7	4.3.3
Type III front handle design	X		3.13.2.2.4.1	4.3.3
Type III rear handle trigger	X		3.13.2.2.4.2a	4.4.6
Type III rear handle design	X		3.13.2.2.4.2b 3.13.2.2.4.2c	4.3.3
Type III light output	X		3.13.2.2.6.1 3.13.2.2.6.2 3.13.2.2.6.3	4.4.8
Enclosure effectiveness	X		3.13.2.3	4.4.10
Weight	X	X	3.14	4.3.3
Identification and switch plates	X	X	3.15	4.3.3
Protection against corrosion	X	X	3.16	4.3.3
Painting	X	X	3.17	4.3.3
Workmanship	X	X	3.18	4.3.3

4.2 First article inspection. First article inspection shall be performed on a multipurpose signal light when a first article sample is required (see 3.1). This inspection shall include the examination of 4.3.3 and the tests of 4.4.1 through 4.4.11.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.3.3 and the tests of 4.4.1, 4.4.2, and 4.4.3.

4.3.1 Lot. For the purpose of sampling, a lot shall be the identical number of signal light kits specified (see 6.2).

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4.3.2 Sampling for examination and tests. A random sample of signaling light kits shall be selected in accordance with table II from each lot. The maximum allowable number of defects shall be as specified (see 6.2).

TABLE II. Sampling for examination and tests.

Number of signal light kits in lot	Number of signal light kits in sample
2 to 10	All
11 to 15	10
16 to 25	13
26 to 40	17
41 to 65	22
66 to 110	28
111 to 180	35
181 to 300	45
301 to 500	55
501 and over	70

4.3.3 Examination. The samples selected in accordance with 4.3.2 shall be examined to determine compliance with the requirements of this specification for design, construction, materials, threaded parts, welding, castings, dimensions, electrical requirements, kitting, the capability and components of the light assembly, the presence and construction of the extension cable, the presence and construction of the power cable, the presence of the instruction manual, the characteristics of the battery or batteries, the compartmentation of the stowage box, the electrical components of the stowage box, weight, marking, protection against corrosion, painting, and workmanship.

4.3.4 Tests. The sample(s) selected in accordance with 4.3.2 shall be subjected to the operation test, the dielectric withstanding voltage test, and the insulation resistance tests of 4.4.1, 4.4.2, and 4.4.3.

4.4 Test procedures.

4.4.1 Operation. The sample(s) shall be checked for operation with both self-contained battery power and the power supply from the stowage box. The light assembly shall be operated at both high and low intensity. The light shall emit steady illumination at both intensities until the trigger switch is depressed. When the trigger switch is depressed, the light shall be interrupted until the switch is released.

4.4.2 Dielectric withstanding voltage. Dielectric withstanding voltage between electrical circuits and between electrical circuits and ground (equipment frame), shall be tested with a closely sinusoidal source of 60 cycles, AC in accordance with MIL-STD-202-301. The root mean square value of the test voltage shall be 1,000 volts. For type I signal lights, the lamp shall be disconnected during this test.

4.4.3 Insulation resistance. The insulation resistance between electrical circuits and between electrical circuits and ground (equipment frame) following the dielectric tests specified in 4.4.2 shall be not less than the value specified in 3.10.3. Insulation resistance shall be measured with a megohmmeter at a voltage not less than 500 volts, DC, at normal room temperature. For type I signal lights, the lamp shall be disconnected during this test.

4.4.4 Shock. The stowage box, complete with component parts (see 3.13.1), shall be securely strapped to the 4C test plate and subjected to a lightweight, grade A, class I, type A shock test in accordance with MIL-DTL-901. The operation test of 4.4.1 shall be performed after completion of the MIL-DTL-901 lightweight shock test to verify the equipment has met the requirements.

4.4.5 Vibration. The stowage box, complete with component parts (see 3.13.1), shall be subjected to type I vibration tests in accordance with MIL-STD-167-1. The vibration test, in accordance with MIL-STD-167-1, shall consist of frequencies up to and including 33 hertz at the table amplitude specified therein. The operation test of

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4.4.1 shall be performed after completion of the MIL-DTL-901 lightweight vibe test to verify the equipment has met the requirements.

4.4.6 Trigger switch life. The trigger switch and mechanism shall be subjected to a life test of 50,000 cycles of operation at the rate of 15 cycles per minute to determine conformance with 3.13.2.1.5.2 or 3.13.2.2.4.2, as applicable. One cycle is defined as one complete movement from the OFF position to the ON position and back to the OFF position.

4.4.7 Magnetic permeability. The magnetic permeability of material shall be determined with an indicator in accordance with ASTM A342, or equal.

4.4.8 Photometrics. Photometric testing shall be required for type II and type III light assemblies only. Testing shall be performed in accordance with ANSI/IES LM-79. Testing shall be performed at both the high (full) and low intensities both on battery power and connected to the stowage box power supply.

4.4.8.1 Type II photometric testing. Testing shall be performed without lenses installed. A candlepower distribution curve shall be obtained to verify that the light assembly meets the beam spread requirements of 3.7. Test data shall also include the power draw, total lumens, and color temperature.

4.4.8.2 Type III photometric testing. Testing shall be performed for white, red, yellow, and green outputs. Candlepower distribution curves shall be obtained to verify that the light assembly meets the beam spread requirements of 3.7. Test data shall also include the power draw, total lumens, color temperature (for white) or color coordinates in the CIE 1931 color space (for red, yellow, and green).

4.4.8.2.1 Type III battery runtime. The ability of the chosen battery to meet the requirement of 3.13.6.2 shall be verified by photometric testing. The total lumens of the type III assembly shall be measured at full brightness on new batteries. The trigger shall be depressed for a minimum of 60 minutes. After 60 minutes, the total lumens shall be measured again.

4.4.9 Continuity of grounding. Electrical resistance shall be measured with a milliohm meter between the ground prong of the power cord male plug and all metal components of a completely connected signal light.

4.4.10 Enclosure effectiveness. The light assembly shall be tested to the spraytight rating in accordance with MIL-STD-108.

4.4.11 Electromagnetic emissions. The light assembly shall be subjected to the RE102 test of MIL-STD-461 while operating on both battery power and from 115 VAC service.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. This kit is designed to provide limited line of sight communications utilizing visual communication methods (Morse code) in accordance with U.S. Navy communications procedures. The requirements herein are designed to ensure signaling is available when other methods fail. The multipurpose light can be used for:

a. Ship to ship signaling.

b. Ship to plane signaling.

c. Ship to shore signaling.

d. Portable spotlight.

e. Covert operation (kit can be modified for infrared capability).

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The lighting industry is transitioning away from incandescent light sources in an effort to reduce power consumption and increase lifetime. The availability of the sealed beam PAR46 lamp of the type I Sym 106.1 signal light will be affected by this trend and will face obsolescence. The addition of type II (LED) and type III (SSL) signal light styles is intended to address future obsolescence in two separate ways, providing flexibility to the procuring activity. The requirements for the type II signal light are intended to minimize the impact to established logistics by utilizing the same housings, cables, batteries, transformer, and filters. The light source is changed from an incandescent lamp to an LED device. The circuitry necessary to work off both AC and DC power sources, interrupt the light output for signaling, and provide high and low intensity outputs are contained entirely within the light assembly. The control of the light source via the trigger and switches on the rear of the light assembly remain the same for the end user, and the new light assembly could be used with the existing stowage box without modification. The requirements for the type III signal light are intended to allow for significant changes to the design of the multipurpose light, taking advantage of new lighting technologies while not being limited by the configuration of the legacy light. The intent is to provide the same basic functionality and light output as a type I device, at a minimum, while allowing for changes to the system architecture such as the operating voltage of the light source, the batteries used, or the use of filters.

Current stock of the Sym 106.1 multipurpose signal light may be used until depleted; future requisitions of the Sym 106.1 multipurpose signal light should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1).

c. The requirement for and the contents of an instruction manual (see 3.13.5).

c. When a battery is required to be furnished (see 3.13.6.1 and 3.13.6.2).

d. Number of signal light kits per sampling lot (see 4.3.1).

e. Maximum allowable number of defects (see 4.3.2).

f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

LED
Light emitting diode
Signaling
Solid state lighting
SSL
Visual communication

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 5850-2025-001)

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